

Hi, my name is Ella Yang, and I'll be going to 11th grade this fall. My project is called Equille, which is a virtual reality horse care and riding simulation. The goal of my project is to let players experience horse care, riding, and show jumping in a safe, realistic, and educational VR environment. It is especially designed to help children, including children with autism, experience the calming structure of horse-related activities without needing access to a real stable or live horses.

I was inspired to create this project because horseback riding can be expensive, risky, and hard to access for many people. Some children may not live near a stable, and others may not feel comfortable around large animals at first. I wanted to use VR to make horseback riding more accessible. I also wanted the game to have a therapeutic side, because horses can teach patience, calmness, routine, and emotional regulation. In my project, the player can care for the horse, build trust, and practice riding skills in a structured digital space. Another reason why I was inspired to create this project was because there was many people with autism at my barn, and they didn't fully understand how to take care of horses and enjoy horseback riding to its fullest, so I wanted to create a game that could teach them that skill and give them a better experience.

Now I will explain how to play the game, first the player will arrive at a virtual stable environment, this is where the horse is located and the first part of the game where they focus on horse care and bonding, players will be able to tack up their horse and groom it. After tacking up the horse, the player will get on the horse and experience the horse walking, trotting, cantering, and jumping. They are able to either take a lesson where they learn all the controls and/or begin a competition after they feel that they are confident and ready.

The whole project was built and tested on unity and the code that was used is called C#. Some features that are included in this game are, a food meter, a thirst meter and a performance meter. (These meters will die down as the horse gets tired from working but could get replenished at its stations.) Players can also set up their own obstacles by first grabbing an obstacle and placing more and more with the A button.

The two major functions in my program that runs my project are the horse movement function and the grooming and bonding function. The horse movement function checks the VR controller input and translates it into riding actions. For example, when both grab buttons are held, the horse moves forward. When the player uses the left grab and left trigger the horse moves to the left, and if the player uses the right grab and right trigger, the horse turns right. Halt is expressed when no buttons are pressed. It also controls the different speed states, including walk (A button), trot (b button), and canter (Y button), and it handles jumping when both joysticks are pushed forward. Another thing to add is when you mount the horse, simply grab the saddle and pull up, and to dismount press the X button.

The second major function is the horse care and bonding function. This function connects the care tasks to the horse's behavior. For example, grooming, feeding, and tacking up increase the horse's trust and readiness. If the player takes care of the horse consistently, the horse performs better during riding and competition. This function supports the main purpose of the game, which is to combine horse care, emotional connection, and riding progression.

One of the most difficult parts of this project was figuring out how to translate real-life riding movement into VR controls. In real horseback riding, the rider uses their hands, legs, balance, and body position, but in VR I had to simplify that into controller inputs. From this project, I learned how important planning is before building a game. I had to define the core mechanics, create a gameplay loop, design the

first level, choose assets, and think about how each feature connects to the purpose of the project. I also learned that a game can be educational and therapeutic while still being fun.

For the next part of the project, I am working on improving the horse behaviors and expanding the playable prototype. I want to add more realistic reactions, such as the horse avoiding the player if the bond level is low, a dirt meter that shows how clean or dirty the horse is, and better movement controls.